

Technical Document #133: Blockchain - Comprehensive Overview

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Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Tokenomics studies the economic systems of cryptocurrency tokens. It covers supply mechanisms, incentive structures, and value capture.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Zero-knowledge proofs allow one party to prove knowledge of information without revealing the information itself. They enhance privacy on blockchains.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Tokenomics studies the economic systems of cryptocurrency tokens.
- Blockchain is a distributed ledger technology that records transactions across many computers.
- Non-fungible tokens (NFTs) represent unique digital assets on the blockchain.